

PPS
Assignment :-

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Class:- ECE 'A'

1Q. Write about Generations of Programming Languages.

Ans:- 1st Generation :- Machine Language (1940-1950s)

This is the lowest level of Programming. It consists entirely of binary code (0s and 1s) that the computer's CPU can execute directly.

→ Characteristics:- Hardware-specific; no translation needed.

→ Pros : Extremely fast and efficient.

→ Cons:- Nearly impossible for human to read (or) debug, if you switched computers, you often had to rewrite the entire program.

2nd Generation:- Assembly Language (1950s)

To make life easier, programmers started using mnemonics instead of binary.

Characteristics:- Requires an Assembler to convert code into the machine language.

Pros:- Easier to read than binary while still offering total control over hardware.

Cons:- Still hardware-specific (Low-level)

3rd Generation : High Level Languages (1950 - Present)

This generation introduced "English Like" syntax, making programming accessible to more people.

These languages are portable, meaning the same code can run on different types of computers.

Ex:- C, C++, Java, Python, COBOL, Fortran.

Characteristics:- Require to translate focus:

→ Characteristics:- Requires a Compiler (or) interpreter to translate the code.

→ Focus:- the logic of the problem rather than the architecture of the machine.

4th Generation :- Very High Level Languages
(1970 - Present)

These are "non-procedural" languages. Instead of telling the computer how to do something step-by-step, you tell it what you want the result to be. They are often used for database management and reports.

Ex: SQL (Structured Query Language), R, MATLAB.

→ Pros : Highly effective for specific tasks, very few lines of code required.

→ Cons:- Less versatile than 3rd Generation for general software development.

5th Generation:- Artificial Intelligence and Natural Language (Present and future) :-

5th Generation are designed to make the computer solve problems without a programmer writing a specific algorithm. They are heavily used in AI, expert systems, and neural networks.

Ex:- Prolog, OPS5, Mercury.

Focus:- Constraint - based programming and logic. The goal is for humans to eventually communicate with computers using normal conversational speech.

Q. Write about Generations of Computers.

Ans:- 1st Generation :- Vacuum Tubes (1940-1954)

These early computers used vacuum tubes for circuitry and magnetic drums for memory. They were massive, often taking up entire rooms, and were incredibly expensive to operate.

→ Technology:- Vacuum tubes

→ Input/output:- Punched cards and paper tapes.

→ Characteristics:- Generated enormous heat, frequent fused out, and could only solve one problem at a time.

Ex:- ENIAC, UNIVAC.

2nd Generation:- Transistors (1956-1963)

The invention of the transistor replaced bulky vacuum tubes. This allowed computers to become smaller, faster, and more energy efficient. This era also saw the move from cryptic binary to symbolic languages.

→ Technology : Transistors

→ Input/output:- Punched cards for inputs, printouts for output.

→ Characteristics:- Faster and more reliable than the 1st gen, but still produced significant heat

→ Ex: IBM 7094, CDC 1604.

3rd Generation

3rd Generation:- Integrated Circuits (1964-1970):-

Instead of individual transistors, scientist figured out how to fit many transistors onto a single silicon chip, called a semiconductor (or) Integrated circuit (IC).

-) Technology:- Integrated circuit (IC).
-) Input/output:- Keyboards and monitors (replacing punched cards).
-) Characteristics:- This way the first time computers became accessible to a mass audience because they were significantly smaller (or) cheaper.

Ex:- IBM 360 series, PDP-8.

4th Generation:- Microprocessors (1971 - present)

This generation is defined by LSI

(Large scale Integration) and VLSI (Very large scale Integration). Thousands of integrated circuits were built onto a single silicon chip - the microprocessors.

Technology:- Microprocessors (CPU's).

Characteristics:- This led to the birth of the Personal Computer (PC). Modern ~~laptops~~ laptops, smartphones and tablets all fall into this category.

Ex:- Apple II, Altair 8800, modern Intel / AMD Ryzen - based PC's.

5th Generation:- Artificial Intelligence (Present and Beyond). :-

The current and future generation is based on parallel processing and Artificial Intelligence (AI). The goal is to develop devices that can respond to natural language and are capable of self-organisation (or "learning").

→ Technology:- VLSI (ultra large size integration) and Quantum Computing.

→ Characteristics:- voice recognition, facial recognition, and complex decision making.

→ Example:- Google Assistant, Siri, Quantum Computer (IBM Q), and Neural Networks.

Q. A person crosses a 600 m long distant street in 5 minutes. what is the speed in km per hour?

Sol:-
$$\text{Speed} = \left[\frac{600}{5 \times 60} \right] \text{ m/sec}$$

$\Rightarrow 2 \text{ m/sec}$

convert it into km/hr.

$\Rightarrow \left[2 \times \frac{18}{5} \right] \text{ km/hr}$

$\Rightarrow 7.2 \text{ km/hr}$

Q. An aeroplane covers a certain distance at a speed of 240 kmph in 5 hrs. To cover the same speed in $1\frac{1}{2}$ hr it must travel at a speed of :-

Sol:-

$\text{Dist} = \text{speed} \times \text{time}$

$\Rightarrow 240 \times 5 \text{ hrs}$

$\Rightarrow 1200 \text{ km}$

$$\text{Speed} = \frac{\text{Dist}}{\text{Time}} \Rightarrow \frac{1200}{(5/3)} \text{ km/hr.}$$

$$\Rightarrow \frac{240}{1200} \times \frac{3}{5} \text{ km/hr}$$

$$\Rightarrow 720 \text{ km}$$

Number systems:-

Decimal number:-

$$5691.9$$

$$\Rightarrow 5 \times 10^3 + 6 \times 10^2 + 9 \times 10^1 + 1 \times 10^0 + 9 \times 10^{-1}$$

Binary Number system:-

$$1101.101$$

$$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1$$

$$\Rightarrow (15.625)_{10}$$

Octal Number system:-

$$5623.421$$

$$5 \times 8^3 + 6 \times 8^2 + 2 \times 8^1 + 3 \times 8^0 + 4 \times 8^{-1} + 2 \times 8^{-2} + 1 \times 8^{-3}$$

$$1 \times 16^3$$

$$\Rightarrow (2990.611328)_{10}$$

Hexa decimal system :-

$$37D.84$$

$$\Rightarrow 3 \times 16^2 + 7 \times 16^1 + D \times 16^0 + 8 \times 16^{-1} + 4 \times 16^{-2}$$

$$\Rightarrow (1021.515625)_{10}$$

Number base conversion :-

Binary to octal conversion

① Convert 10101101.0111 to octal

$$\begin{array}{ccccccc} 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & . & 0 & 1 & 1 & 1 \\ \leftarrow & & & & & & & & & & & & & \\ 2 & 5 & & 5 & & & 3 & 4 & & & & & & \end{array}$$

$$\Rightarrow (55.34)_8$$

②

octal to binary conversion

$$(125.62)_8 \text{ to binary}$$

$$001\ 010\ 101\ .\ 110\ 010$$

$$(125.62)_8 = (1010101.11001)_2$$

3

Binary to Hexadecimal

Convert 1101101110.1001101

00110110110.10011013
 ← 3 ← 6 ← E → 9 → A

octal to Hexa.

(615.25)₈

00110001101.01010100
 ← 1 ← 8 ← D → 5 → 4

(18D.54)₁₆

5

Hexa to octal

(B66.AE)₁₆

001011110001100110.10101113
 ← 3 ← 6 ← 4 ← 6 → 5 → 3 → 6

(136146.536)₈

⑥

Convert 12.125 decimal to binary

Step-1:- $12 \times 2^{-1} = 6 - 0$

$6 \times 2^{-1} = 3 - 0$

$3 \times 2^{-1} = 1 - 1$

$1 \times 2^{-1} = 1 - 0$

Step-2:- $0.125 \times 2 = 0.25 - 0$

$0.25 \times 2 = 0.50 - 0$

$0.50 \times 2 = 1.00 - 1$

$(12)_{10} = (0100)_2$

$(0.125)_{10} = (0.001)_2$

Binary to decimal:-

$(10101)_2$

1 0 1 0 1

$1 \times 2^0 = 1$
 $0 \times 2^1 = 0$
 $1 \times 2^2 = 4$
 $0 \times 2^3 = 0$
 $1 \times 2^4 = 16$

↓
21

$(10101)_2 = (21)_{10}$

Q.

Flowchart :-



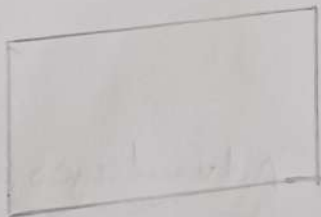
oval

Start/stop (end).



Parallelogram

Input/output



Rectangle

process / instruction



diamond

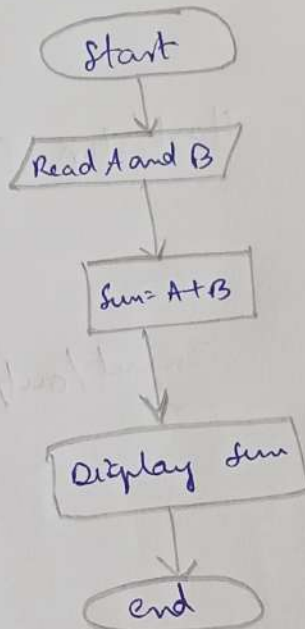
Decision (Yes/No).



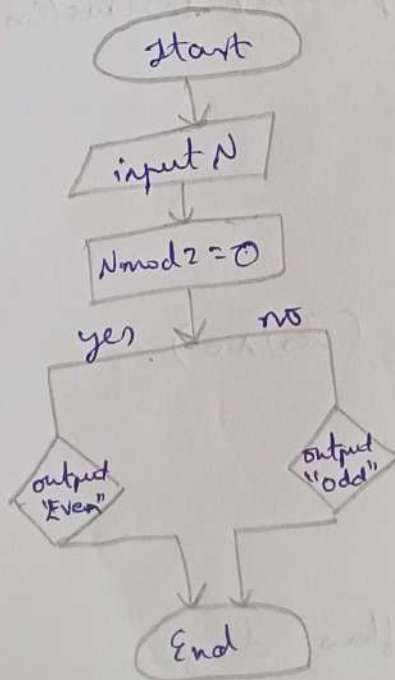
Arrow

shows flow diagram

Ex:- 1. Sum of two numbers:-



2.



Advantages:-

- Simple to understand
- Helps to debugging
- Easy to convert.